

2020 HKDSE Physics

Paper 1B — Structured Questions

Editable transcription of all questions and formulae. Only necessary, non-duplicated original diagrams are inserted. Each official answer is followed by a verified concise full-mark solution in English, with editable subscripts and superscripts.

Editable transcription with official answers and verified concise full-mark reasoning in English.

Official answers are placed directly below each question.

Question 1

In a restaurant, ‘wontons in soup’ is prepared by putting 5 pieces of cooked wonton at 4°C into a bowl with 0.60 kg of soup at 96°C . Given: average mass of each piece of wonton = 0.02 kg specific heat capacity of wonton = $3300\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$ specific heat capacity of soup = $4200\text{ J kg}^{-1}\text{ }^{\circ}\text{C}^{-1}$

- (a) Find the final temperature of the mixture. Assume that the heat capacity of the bowl and heat loss to the surroundings are negligible. (2 marks)
- (b) The soup in (a) is taken from a metallic container of heat capacity $2000\text{ J }^{\circ}\text{C}^{-1}$ containing 16 kg of soup maintained at 96°C by an immersion heater.
 - (i) Why does energy have to be supplied by the heater to keep the soup at 96°C ? (1 mark)
 - (ii) A student used the following method to find the heater's operating power P : remove the heater from the container and record the temperature of the 16 kg of soup after 10 minutes. It is found that the temperature has dropped 9°C . Estimate P . (3 marks)
 - (iii) If the student repeats the measurement after another 10 minutes, would the corresponding temperature drop be larger than, equal to or smaller than 9°C ? Explain. (2 marks)

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Solution	Marks	Remarks
1. (a) $5 \times 0.02 \times 3300 \times (T - 4) = 0.60 \times 4200 \times (96 - T)$ $T = 85.347368 \text{ }^{\circ}\text{C} \approx 85.3 \text{ }^{\circ}\text{C}$	1M 1A	Accept $85.0 \text{ }^{\circ}\text{C}$ to $85.4 \text{ }^{\circ}\text{C}$ [Marking Remark (MR): temperature difference must be correct & accept missing '5' – 1 M]
(b) (i) To compensate for / balance the heat loss (of the container with soup) to the surroundings.	1A	[MR: Key words - heat loss to the surroundings (from container or through evaporation)]
(ii) $P \times 10 \times 60 = 2000 \times 9 + 16 \times 4200 \times 9$ $P = 1038 \text{ W} \approx 1040 \text{ W}$	1M+1M 1A	Assume : Power of the heater = Rate of heat loss (of the container with soup) to the surroundings
(iii) Smaller than $9 \text{ }^{\circ}\text{C}$. As the temperature of (the container with) soup drops, the rate of heat loss to the surroundings becomes lower due to a smaller temperature difference (w.r.t. the surroundings).	1A 1A	[MR: $P \times t$ ✓ + 1 term ✓ - 1M $P \times t$ ✓ + 2 term ✓ - 2M $P \times t$: accept $P \times 10$ ΔT : accept $(96 - 9)$] [2 A marks are independent Key concept: smaller temperature difference]

(a) At equilibrium, heat lost by the soup equals heat gained by the wontons: $5(0.020)(3300)(T-4) = 0.60(4200)(96-T)$. Solving gives $T = 85.3 \text{ }^{\circ}\text{C}$.

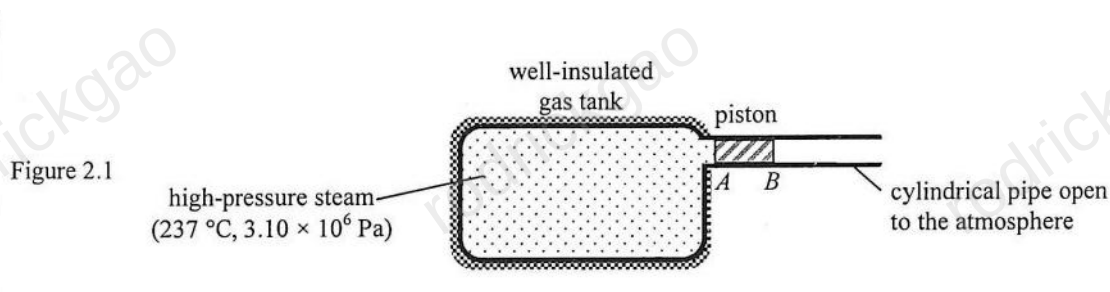
(b)(i) The heater supplies energy at the same rate as the soup loses energy to the surroundings, so the temperature remains constant.

(ii) Over 10 min, $P(10 \times 60) = 2000(9) + 16(4200)(9)$, giving $P = 1.038 \times 10^3 \text{ W} \approx 1.04 \text{ kW}$.

(iii) The temperature fall is less than $9 \text{ }^{\circ}\text{C}$ because, as the soup cools, its temperature difference from the surroundings and hence its rate of heat loss both decrease.

Question 2

Figure 2.1 shows a large gas tank connected with a cylindrical pipe open to the atmosphere. The pipe is fitted with a smooth piston AB. This well-insulated gas tank is filled with high-pressure steam at a temperature of $237 \text{ }^{\circ}\text{C}$ under a pressure of $3.10 \times 10^6 \text{ Pa}$ while the movable piston is held stationary by a force F_p . Given: atmospheric pressure = $1.0 \times 10^5 \text{ Pa}$



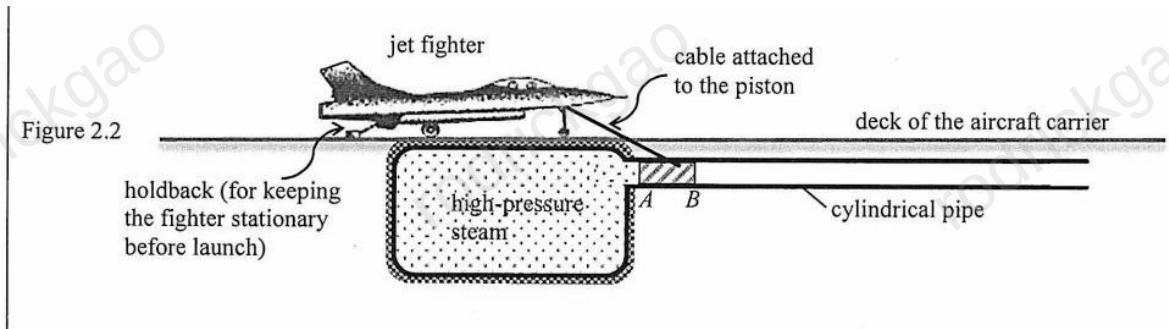
(a) (i) On Figure 2.1 indicate the force F_p . (1 mark)

(ii) By considering the force acting on the piston due to the difference in pressure, find the value of F_p . The piston has a cross-sectional area of 0.67 m^2 . (2 marks)

(iii) Estimate the volume of the gas tank which contains 570 kg steam. You may treat the steam as an ideal gas.

Given: mass of one mole of steam = 0.018 kg. (3 marks)

(b) This set-up can be used as a 'steam catapult' to launch jet fighters from an aircraft carrier. A jet fighter in position to be launched is connected to the piston via an inextensible cable as shown in Figure 2.2. When the holdback behind the jet fighter is released, the high-pressure steam in the gas tank expands and pushes the piston, which in turn helps to accelerate the jet fighter.



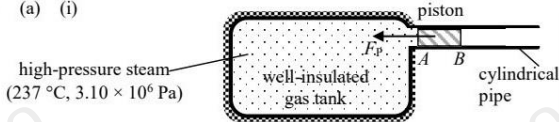
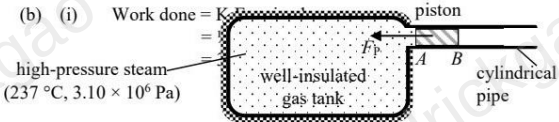
In a trial run of the catapult, a jet fighter (with its engine shut down) acquires a final speed of 54 m s^{-1} 1.5 s after running a distance horizontally on the deck. The mass of the jet fighter is $2.6 \times 10^4 \text{ kg}$.

- Find the work done by the net force on the jet fighter during launch. (2 marks)
- Calculate the average acceleration of the jet fighter during launch. (2 marks)
- State whether the acceleration of the jet fighter is increasing, decreasing or uniform during launch. Explain your answer. (3 marks)

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Solution	Marks	Remarks
2. (a) (i) 	1A	F_p correctly marked F_p should be in contact with the piston. Accept: no label of the force
(ii) $F_p = (3.10 \times 10^6 - 1.0 \times 10^5) \times 0.67$ $= 2010000 \text{ N} = 2.01 \times 10^6 \text{ N}$	1M 1A 2	Accept: $F_p = (3.10 \times 10^6) \times 0.67$ Accept $2.00 \times 10^6 \text{ N}$ to $2.01 \times 10^6 \text{ N}$
(iii) $pV = nRT \Rightarrow V = \frac{nRT}{p}$ $V = \frac{(570/0.018)(8.31)(237 + 273)}{3.10 \times 10^6}$ $= 43.292419 \text{ m}^3 \approx 43.3 \text{ m}^3$	1M+1M 1A 3	1 st 1M find n (570/0.018) 2 nd 1M $pV = nRT$, accept $T = 237$ Accept 43.0 m^3 to 43.3 m^3
(b) (i) Work done = K.E. 	1M 1A 2	Accept 37.9 MJ to 38.0 MJ Accept: $a = 54/1.5 = 36 \text{ (m s}^{-2}\text{)}$ $F = (2.6 \times 10^4) \times 36$ $= 936000 \text{ (N)}$ $s = 54^2 / (2 \times 36) = 40.5 \text{ (m)}$ [or $v = at/2 = 36(1.5)/2$ $= 27 \text{ (m s}^{-1}\text{)}$ $W = Fs = 936000 \times 40.5$ [or $W = Fvt = 936000 \times 27 \times 1.5$ $= 37908000 \text{ (J)}$ [1M+1A] If $s = vt$ [0M 0A]
(ii) Average acceleration $a = \frac{v-u}{t} = \frac{54-0}{1.5}$ $= 36 \text{ m s}^{-2}$	1M 1A 2	1 st 1A for acceleration decreasing 2 nd 1A for volume increases/ length increases $\Rightarrow$ pressure decreases
(iii) Acceleration is decreasing (i.e. maximum at first). (According to kinetic theory, once the steam expands, i.e. volume increases, its pressure decreases, thus the (pressure) force acting on the piston at A decreases, hence a smaller acceleration.	1A 1A 1A 3	Alternative method: The acceleration of the jet fighter is decreasing, 1A As the air resistance (f) acts on the jet fighter increases with the speed. 1A The net force ($ma = F - f$) is decreasing. 1A

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(a)(i) Mark F_p horizontally to the left on the piston, toward the gas tank, so that it balances the net rightward pressure force.

(ii) $F_p = (3.10 \times 10^6 - 1.00 \times 10^5)(0.67) = 2.01 \times 10^6 \text{ N}$.

(iii) The amount of steam is $n = 570/0.018 \text{ mol}$. Hence $V = nRT/p = (570/0.018)(8.31)(510)/(3.10 \times 10^6) = 43.3 \text{ m}^3$.

(b)(i) The work done becomes kinetic energy: $W = \frac{1}{2}mv^2 = \frac{1}{2}(2.6 \times 10^4)(54^2) = 3.79 \times 10^7 \text{ J} = 37.9 \text{ MJ}$.

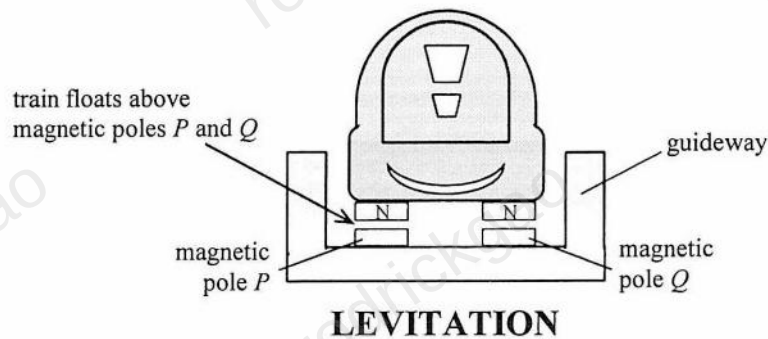
(ii) The average acceleration is $a = \Delta v / \Delta t = 54 / 1.5 = 36 \text{ m s}^{-2}$.

(iii) The acceleration decreases because the expanding steam occupies a larger volume and its pressure, hence the resultant driving force, decreases.

Question 3

Read the following passage about a magnetically levitated (maglev) train and answer the questions that follow.

‘A maglev train car is just a box with magnets on the four corners,’ says Jesse Powell, the son of the maglev train inventor. The electromagnets employed have superconducting coils (i.e. coils with extremely low resistance). They therefore can generate magnetic fields 10 times stronger than ordinary electromagnets, enough to levitate and propel a train.



Two sets of magnetic fields are set up for different functions. One is to make the train float a few centimetres above magnetic poles P and Q as shown, while the other is a propulsion system run by an alternating current. The floating design enables a smooth movement of the train. Even when the train travels up to 600 km per hour, passengers inside experience less vibration than travelling on traditional trains.

- Explain why electromagnets employing superconducting coils can produce much stronger magnetic fields. (2 marks)
- State the polarities of magnetic poles P and Q and explain how this arrangement enables the train to float. (2 marks)
- Referring to the resistive forces experienced by the train, explain why a maglev train ride is (i) smoother and (ii) faster. (2 marks)

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Solution	Marks	Remarks																				
3. (a) With superconductor / extremely low or nearly zero resistance, a much larger current flows in the coils producing a stronger magnetic field (with less heat loss due to the current flow).	1A 1A 2	[MR: low R + large I – 2A marks large I only – 1 A mark low R only – 0 A mark]																				
(b) P and Q : N (north pole) Repulsive force between like poles	1A 1A 2	<table><tr><th colspan="5">Poles</th></tr><tr><th></th><th>2N</th><th>2S/S+N</th><th>正</th><th>NA</th></tr><tr><td>Reason</td><td>✓</td><td>2</td><td>0</td><td>1</td></tr><tr><td></td><td>✗/NA</td><td>1</td><td>0</td><td>0</td></tr></table>	Poles						2N	2S/S+N	正	NA	Reason	✓	2	0	1		✗/NA	1	0	0
Poles																						
	2N	2S/S+N	正	NA																		
Reason	✓	2	0	1																		
	✗/NA	1	0	0																		
(c) (i) As the train is not in contact with the rail, <u>there is no friction / interaction between the train and the rail</u> , thus smoother due to less vibration.	1A	NA – Not answered																				
(ii) There is no friction / interaction between the train and the rail , and the propulsion of the train only needs to work against air resistance, therefore much faster.	1A 2																					

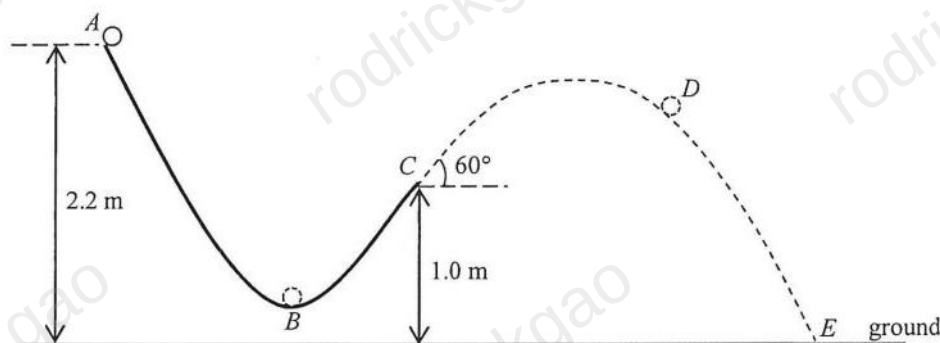
(a) A superconducting coil has essentially zero resistance, so it can carry a much larger current with negligible heating and produce a stronger magnetic field.

(b) P and Q must be like poles, both north poles in the shown arrangement, so their repulsion provides magnetic lift.

(c)(i) The train does not touch the rail, so contact friction and vibration are greatly reduced and the ride is smoother.

(ii) With contact friction removed, mainly air resistance opposes the motion, allowing a higher operating speed.

Question 4



A small sphere is released from rest at point A and runs along a smooth track ABC as shown in Figure 4.1. The track around the lowest point B is approximately circular in shape. The sphere leaves the track at point C where the track makes an angle of 60° with the horizontal. It finally reaches point E on the ground. Neglect air resistance. ($g = 9.81 \text{ m s}^{-2}$)

(a) Arrange the speeds of the sphere at points A, B, C and D respectively in descending order. (1 mark)

(b) On Figure 4.1, use arrows to indicate the acceleration of the sphere, if any, at point B and at point D respectively. (2 marks)

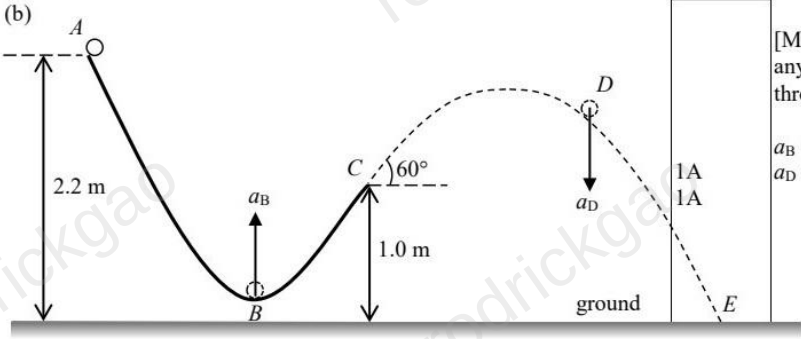
(c) (i) Describe the energy conversion of the sphere when it goes along the track ABC. (2 marks)

- (ii) Hence find the speed of the sphere at point C. (2 marks)
- (iii) If the horizontal distance between points C and E is 2.55 m, calculate the time of flight of the sphere before reaching point E. (3 marks)

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Solution	Marks	Remarks
4. (a) $v_B > v_C > v_D > v_A$	1A	[MR: Accept: $v_A < v_D < v_C < v_B$ Accept: B, C, D, A NOT accept: A, D, C, B]
(b)	1	
	1A 1A	[MR: Accept the arrows drawn at anywhere on the vertical lines passing through the object at positions B & D] a_B correctly marked a_D correctly marked
(c) (i) Gravitational potential energy is changed to kinetic energy from A to B, and some kinetic energy is changed back to gravitational potential energy from B to C.	1A 1A	[MR: 2 marks : All correct and complete: PE → KE & some KE → PE 1 mark : (i) PE → KE → PE or (ii) Correctly describe only between 2 points (A → B / B → C / A → C)]
(ii) $\frac{1}{2}mv^2 = mgh$ $v^2 = 2(9.81)(2.2 - 1.0)$ $v = 4.852216 \text{ m s}^{-1} \approx 4.85 \text{ m s}^{-1}$	1M 1A	[MR: h must be correct for 1M Method using $v^2 - u^2 = 2as$ etc. is NOT accepted] For $g = 10 \text{ m s}^{-2}$, $v = 4.89898 \text{ m s}^{-1} \approx 4.90 \text{ m s}^{-1}$ Accept 4.85 m s^{-1} to 4.90 m s^{-1}
(iii) Horizontal speed (at C) = $4.85 \times \cos 60^\circ$ $= 2.426108 \text{ m s}^{-1} \approx 2.43 \text{ m s}^{-1}$ Distance = $2.43 \times t = 2.55 \text{ m}$ $t = 1.051066 \text{ s} \approx 1.05 \text{ s}$ Or Vertical speed (at C) = $4.85 \times \sin 60^\circ$ $= 4.2002 \text{ m s}^{-1} \approx 4.20 \text{ m s}^{-1}$ $-1.0 = 4.20 \times t + \frac{1}{2}(-9.81)t^2$ $t = 1.051066 \text{ s} \approx 1.05 \text{ s}$	1M 1M 1A	e.c.f. from (c)(ii) For $g = 10 \text{ m s}^{-2}$, Horizontal speed $= 2.44949 \text{ m s}^{-1} \approx 2.45 \text{ m s}^{-1}$ Vertical speed $\approx 4.24 \text{ m s}^{-1}$ and $t = 1.041033 \text{ s} \approx 1.04 \text{ s}$ Accept 1.04 s to 1.10 s [MR: 1 st M for correct expression for initial horizontal / vertical speed 2 nd M for correct method (equation) to find t by considering horizontal motion/ vertical motion
	3	

(a) From energy and height, the speed order is $v_B > v_C > v_D > v_A$.

(b) At B the centripetal acceleration points upward toward the centre of curvature; at D it points

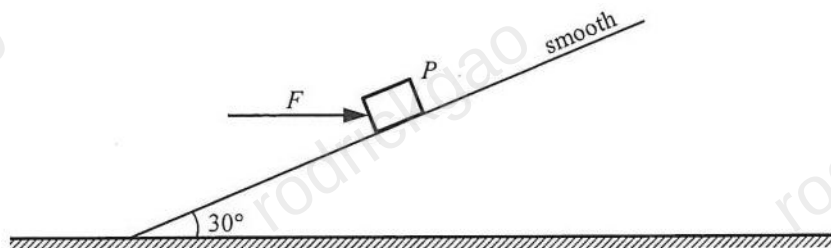
downward.

(c)(i) From A to B, gravitational potential energy is converted to kinetic energy. From B to C, part of the kinetic energy is converted back to gravitational potential energy.

(ii) $\frac{1}{2}mv^2 = mg(2.2-1.0)$, so $v = \sqrt{2g(1.2)} = 4.85 \text{ m s}^{-1}$.

(iii) The horizontal component is $4.85 \cos 60^\circ = 2.43 \text{ m s}^{-1}$. Therefore $t = 2.55/2.43 = 1.05 \text{ s}$.

Question 5



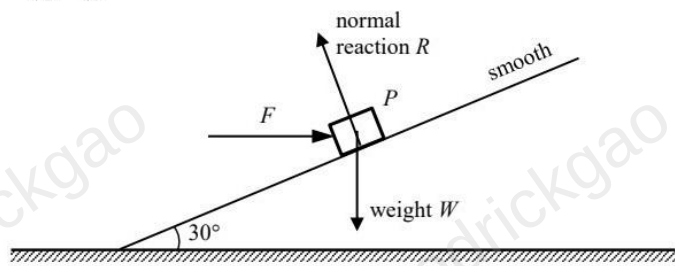
A block P of mass 10 kg is kept stationary on a smooth incline by a horizontal force F as shown in Figure 5.1. The incline makes an angle of 30° with the horizontal. ($g = 9.81 \text{ m s}^{-2}$)

- (a) (i) On Figure 5.1, indicate and label all other forces acting on P. (2 marks)
- (ii) Find the magnitudes of the force F and the force exerted by the block on the incline respectively. (3 marks)
- (b) Now F is removed and air resistance is neglected.
 - (i) What is the magnitude of the acceleration of the block? (1 mark)
 - (ii) Explain whether the force exerted by the block on the incline would increase, decrease or remain unchanged when compared with that in (a)(ii). (2 marks)

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Solution	Marks	Remarks																					
5. (a) (i) 	1A 1A	Forces R (or N) and W (or Mg) correctly indicated and labelled. [MR: 1A for weight with correct label (W/Mg /weight/gravity) 1A for normal reaction with correct label (R/N /normal force/contact force/reaction) 1A only: if both W and R drawn (without/wrong label) Withhold 1A for additional force drawn (more than 2 forces) - W components added with correct labels (ignored) - W components added without correct labels (treated as additional force drawn)																					
	2																						
(ii) $R \cos 30^\circ = W = Mg$ and $R \sin 30^\circ = F$ $F = Mg \times \tan 30^\circ = 56.63806 \text{ N} \approx 56.6 \text{ N}$ $N = R = \frac{Mg}{\cos 30^\circ} = 113.27612 \text{ N} \approx 113 \text{ N}$ <u>Or</u> $R = W \cos 30^\circ + F \sin 30^\circ$ and $W \sin 30^\circ = F \cos 30^\circ$	1M 1A 1A (1M)	$M = 10 \text{ kg}$; $Mg = 98.1 \text{ N}$ [MR: 1M for both equations are correct] For $g = 10 \text{ m s}^{-2}$, $57.735027 \text{ N} \approx 57.7 \text{ N}$ $115.470054 \text{ N} \approx 115 \text{ N}$ [MR: 1M for both equations are correct]																					
	3																						
(b) (i) $g \sin \theta = 9.81 \sin 30^\circ = 4.905 \text{ m s}^{-2} \approx 4.91 \text{ m s}^{-2}$	1A	5 m s^{-2} for $g = 10 \text{ m s}^{-2}$,																					
	1																						
(ii) Decrease as the component of F perpendicular to the incline no longer acts on the block / incline <u>Or</u> as F is removed, the force pressing the block / incline would decrease (only the weight's component left)	1A 1A	<table><tr><th colspan="2" rowspan="2"></th><th colspan="3">Answer</th></tr><tr><th>De</th><th>In / Un</th><th>NA</th></tr><tr><td rowspan="3">Reason</td><td>✓</td><td>2</td><td>0</td><td>1</td></tr><tr><td>✗ / NA</td><td>0</td><td>0</td><td>0</td></tr><tr><td>Not complete</td><td>1</td><td>0</td><td>0</td></tr></table>			Answer			De	In / Un	NA	Reason	✓	2	0	1	✗ / NA	0	0	0	Not complete	1	0	0
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	2	NA – Not answered / Un - Unchanged																					

(a)(i) Add the weight mg vertically downward and the normal reaction R perpendicular to the smooth plane.

(ii) Resolving vertically and horizontally gives $R \cos 30^\circ = mg$ and $R \sin 30^\circ = F$. Hence $F = mg \tan 30^\circ = 56.6 \text{ N}$ and $R = mg / \cos 30^\circ = 113 \text{ N}$.

(b)(i) After the horizontal force is removed, $a = g \sin 30^\circ = 4.91 \text{ m s}^{-2}$ down the plane.

(ii) The normal reaction decreases because the removed horizontal force no longer has a component pressing the block into the plane.

Question 6

The set-up in Figure 6.1 is to find the speed of sound in air. Two identical microphones A and B are placed along a bench top. A digital timer starts and stops timing using the respective microphones to feed signals to the START and STOP terminals of the timer.

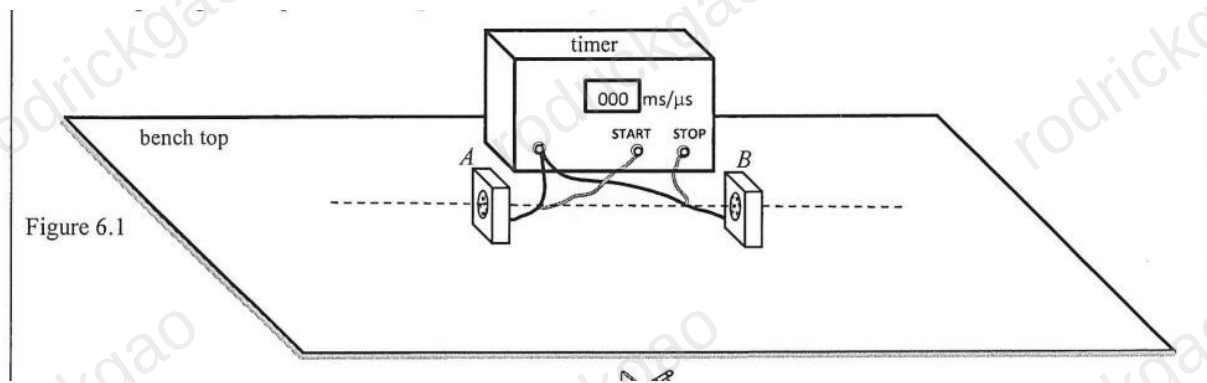


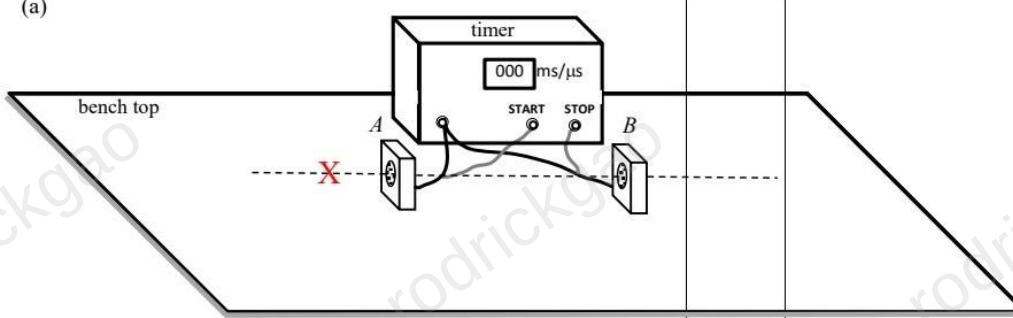
Figure 6.1

- (a) You are given a hammer and a metal plate. Use 'X' to indicate a suitable location on Figure 6.1 where the hammer should hit the plate so as to generate a sharp loud sound to be received by the microphones in this experiment. State an additional piece of apparatus needed and the measurements to be made in this experiment. (3 marks)
- (b) The separation between A and B is set at 0.280 m. The experiment is repeated to obtain a few readings of the timer as follows: 801 μs , 838 μs , 539 μs , 821 μs .
- (i) Find the speed of sound in air. Show how you would treat the data obtained in the calculation. (3 marks)
- (ii) Suggest one adjustment to the experimental setting so as to obtain a more accurate result. (1 mark)

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Solution	Marks	Remarks
6. (a)  Correct location of 'X' (close to axis and on the side of the START microphone). Use a <u>metre rule</u> to measure the distance / separation D between the two microphones. Record the timer reading, which is the time interval Δt for the sound to travel from one microphone (START) to the other microphone (STOP), i.e. A to B.	1A 1A 1A 3	Accept ruler Any ONE of the measurements
(b) (i) Discard $539 \mu\text{s}$, $\Delta t = \frac{801 + 838 + 821}{3} = 820 \mu\text{s}$ The speed of sound in air $v = \frac{D}{\Delta t}$ $v = \frac{0.280}{820 \times 10^{-6}}$ $= 341.463415 \text{ m s}^{-1} \approx 341 \text{ m s}^{-1}$ (ii) Increase separation D between the microphones.	1M 1A 1A 3	1M for taking the average value of Δt and quoting the formula $v = \frac{D}{\Delta t}$ to calculate the speed of sound, where D is the separation between the microphones. Accept 340 m s^{-1} to 342 m s^{-1}

(a) Strike the metal plate close to the START microphone A, with the impact point on the line of the microphones. Measure the microphone separation D and the timer delay Δt , then calculate $v = D/\Delta t$.

(b)(i) The $539 \mu\text{s}$ result is anomalous and should be discarded. The mean of the other readings is $(801+838+821)/3 = 820 \mu\text{s}$, so $v = 0.280/(820 \times 10^{-6}) = 341 \text{ m s}^{-1}$.

(ii) Increase the separation D so that the timing interval is longer and the fractional timing uncertainty is smaller.

Question 7

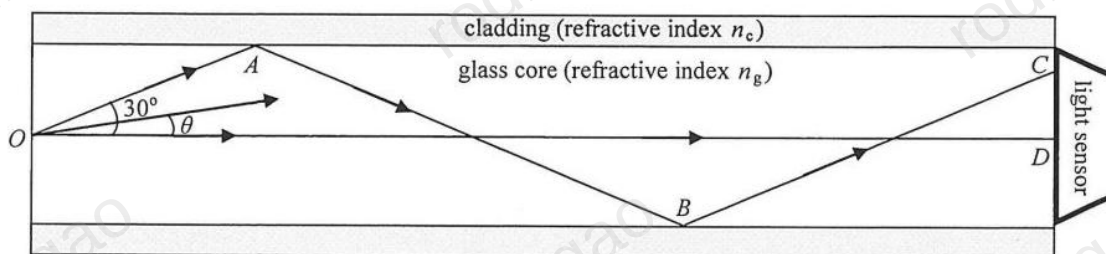
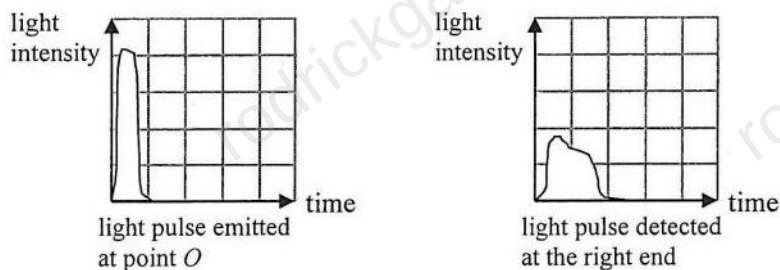


Figure 7.1

Figure 7.1 shows an optical fibre which consists of a cylindrical glass core of refractive index n_g enclosed by a transparent cladding of refractive index n_c . As shown in Figure 7.1, a point light source at O emits monochromatic light in all directions. Inside the fibre, light can reach the right end through many different paths making angles θ with the axis OD . Two of these paths, OD and $OABC$, have been drawn for reference. Light ray OA makes an angle of 30° with the axis OD and is incident at the core-cladding boundary at A with an angle of incidence i_A .

- (a) (i) Find i_A . (1 mark)
 - (ii) If i_A is just greater than the critical angle of that boundary, estimate n_g/n_c . (2 marks)
 - (iii) What phenomenon occurs at point A ? State the condition that needs to be satisfied by θ such that this phenomenon fails to occur. (2 marks)
- (b) A narrow monochromatic light pulse (i.e. of a short duration) emitted at point O propagates with its energy within $\theta = \pm 30^\circ$ towards a light sensor located at the right end of the optical fibre. The respective emitted and detected light pulses are represented in Figure 7.2 using the same scales.



- (i) Explain why the light pulse detected is broader (i.e. of a longer duration) and has lower intensity. Assume that the loss of energy of the light pulse due to absorption by glass is negligible. (2 marks)
- (ii) An engineer suggests changing the refractive index n_c of the cladding in order to reduce the width of the light pulse received. Should n_c be increased or decreased, or will a change in n_c have no effect on the pulse width? Explain your choice. (2 marks)

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Solution	Marks	Remarks
7. (a) (i) Angle of incidence at A, $i_A = 90^\circ - 30^\circ = 60^\circ$	1A	
	1	
(ii) $n_g \sin c = n_c \sin 90^\circ$	1M	1M for correct equation with angle of refraction in the cladding = 90°
$\Rightarrow \frac{n_g}{n_c} = \frac{1}{\sin c} \geq \frac{1}{\sin 60^\circ} = 1.1547005 \approx 1.15$	1A	Accept 1.2 or $\frac{2}{\sqrt{3}}$
	2	
(iii) Total internal reflection. For light rays from O with $\theta > 30^\circ$, they will fail to undergo total internal reflection.	1A 1A	Correct spelling Accept $\theta \geq 30^\circ$
	2	
(b) (i) Some light / energy (of the narrow light pulse) taking the shortest path (i.e. OD) arrives first while the rest of the energy taking longer paths arrives later. Therefore the pulse is broader and the height of the pulse is lower (smaller intensity).	1A 1A	1A for 'taking different paths' 1A for 'different arrival time'
	2	NOT accept: Some of the energy (going at angles larger than 30° from the axis) would be lost due to leakage, thus lower intensity.
(ii) The refractive index of cladding should be increased, as $\frac{n_g}{n_c} = \frac{1}{\sin c}$, by making $c / \sin c$ larger, only those light rays close to the axis / within a smaller θ are totally internal reflected, (such that $\frac{n_g}{n_c}$ getting closer to 1).	1A 1A	0 A for the explanation if the choice for the change of n_c is INCORRECT.
	2	Note: More light rays will escape the optical fibre for a larger n_c .

(a)(i) The angle of incidence at A is $i_A = 90^\circ - 30^\circ = 60^\circ$.(ii) At the limiting ray, $c \approx 60^\circ$, so $n_g/n_c = 1/\sin c \approx 1.15$.(iii) Total internal reflection keeps the light in the core. Rays with $\theta > 30^\circ$ do not meet the total-internal-reflection condition.

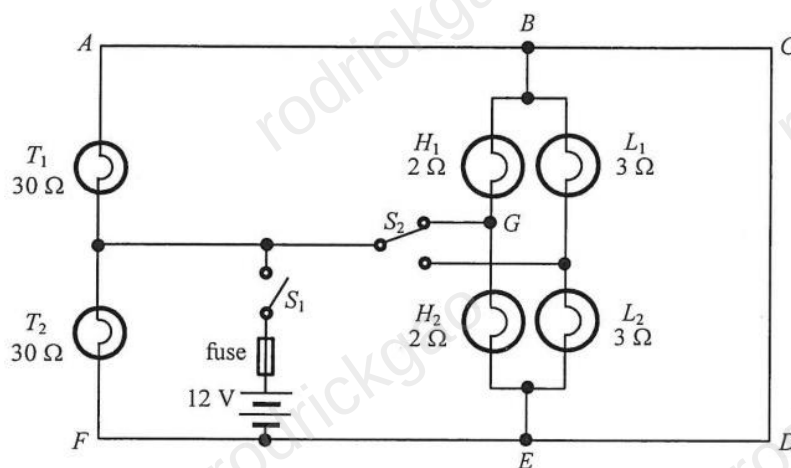
(b)(i) Different rays follow different path lengths and arrive at different times, broadening the pulse. The same pulse energy is spread over a longer time, so the peak intensity decreases.

(ii) Increase n_c . This reduces the core-to-cladding index ratio and admits only rays closer to the axis, reducing the range of path lengths and arrival times.

Question 8

Figure 8.1 shows a simplified circuit of the lighting system of a car. Each of the taillights (T_1 , T_2), high-beam headlights (H_1 , H_2) and low-beam headlights (L_1 , L_2) has resistance $30\ \Omega$, $2\ \Omega$ and $3\ \Omega$ respectively. The internal resistance of the 12 V battery and the resistance of the fuse are negligible. When switch S_1 is closed and switch S_2 is set at the position shown in Figure 8.1, only T_1 and T_2 as well as H_1 and H_2 are lit. The current drawn from the battery is at a maximum in this setting.

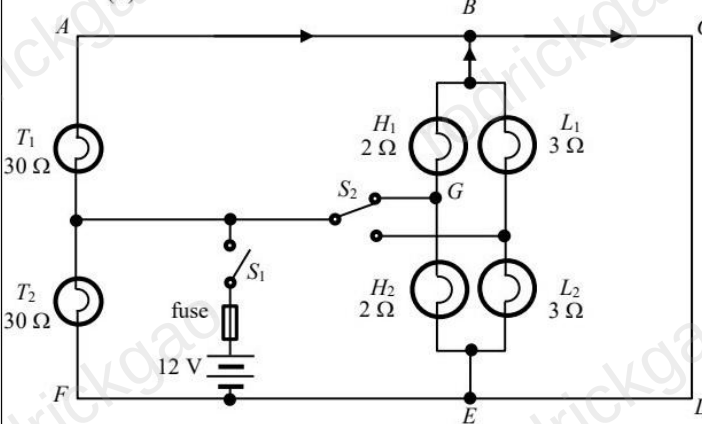
Figure 8.1



(a) Explain why L_1 and L_2 are not lit. (1 mark)

- (b) (i) What is the potential difference across taillight T_1 ? (1 mark)
- (ii) Indicate on Figure 8.1 the direction of current in each of the branches AB, GB and BC. Which branch carries the largest current? (3 marks)
- (c) Calculate the power delivered by the battery and show that the equivalent resistance of the circuit is slightly less than $1\ \Omega$ in this setting. (4 marks)
- (d) Based on your answer in (c), explain whether a fuse rating of $15\ \text{A}$ is suitable for this circuit or not. (2 marks)

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Solution	Marks	Remarks
8. (a) Because L_1 and L_2 (connected in series) are <u>shorted by BCDE</u>	1A	Accept p.d. across each low-beam headlight (L_1 or L_2) is 0.
	1	
(b) (i) p.d. across $T_2 = 12\text{ V}$	1A	
	1	
(ii)  largest current in branch BC	1A 1A 1A 3	Any two currents correctly marked. All currents correct.
(c) Power supplied $P = 2 \times \left(\frac{12^2}{30} + \frac{12^2}{2} \right) = 153.6\text{ W} = 154\text{ W}$ (3 sig. fig.) $\frac{V^2}{R_{eq}} = \frac{12^2}{R_{eq}} = 153.6$ (or $\frac{1}{R_{eq}} = \frac{1}{30} + \frac{1}{30} + \frac{1}{2} + \frac{1}{2}$) $R_{eq} = 0.9375\ \Omega \approx 0.938\ \Omega \approx 1\ \Omega$ (3 sig. fig.)	1M 1A 1M 1A 4	1M for same voltage across each bulb (T_1 , T_2 , H_1 and H_2) e.c.f. from (b)(i) 1M for all bulbs (T_1 , T_2 , H_1 and H_2) in parallel
(d) max. current $I_{max} = \frac{V}{R_{eq}} = \frac{12}{0.9375} = 12.8\text{ A}$ (or $I_{max} = \frac{P}{V} = \frac{153.6}{12} = 12.8\text{ A}$) Since the rating is slightly higher than the max. current (when high-beam headlights and taillights are switched on), it is suitable.	1M 1A 2	e.c.f. from (c) 1A for 12.8 A and correct conclusion

(a) The branch containing L_1 and L_2 is bypassed by the low-resistance path BCDE, so the potential difference across those lamps is zero.

(b)(i) The potential difference across T_2 is 12 V.

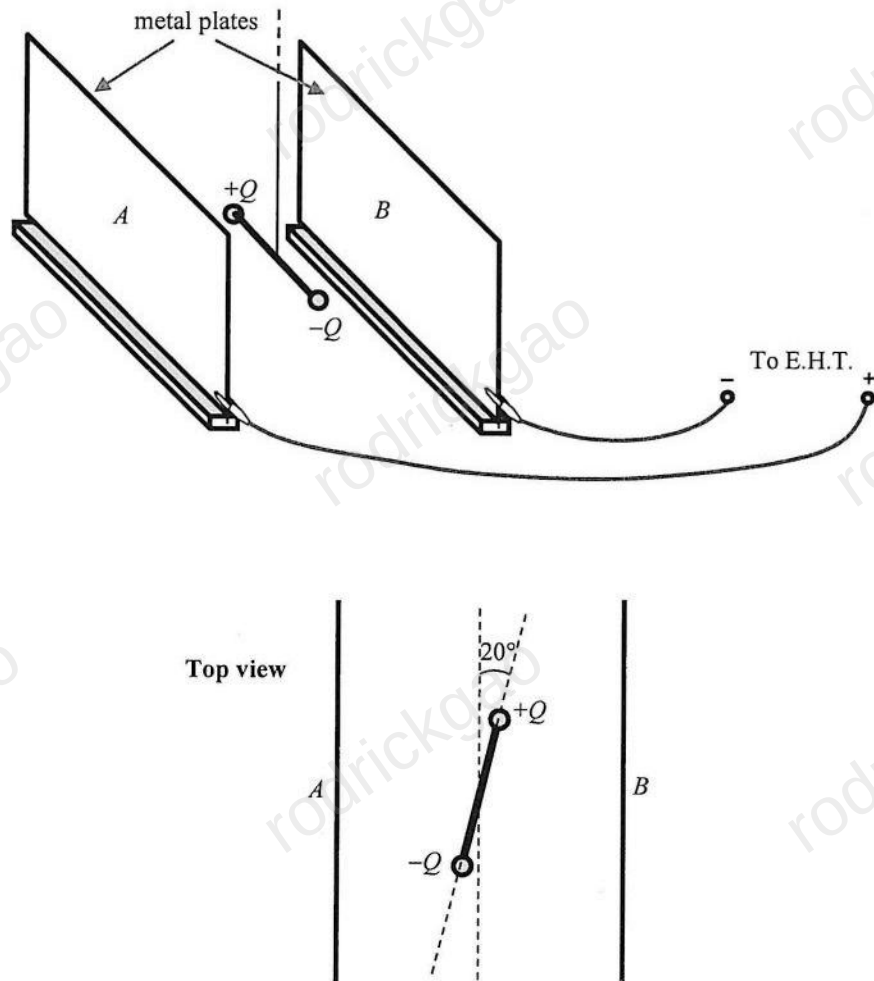
(ii) Conventional current flows from A to B in branch AB, from G to B in branch GB, and from B to C in branch BC. Branch BC carries the largest current.

(c) $P = 2[12^2/30 + 12^2/2] = 153.6\text{ W} \approx 154\text{ W}$. Also $1/R_{eq} = 1/30 + 1/30 + 1/2 + 1/2$, so $R_{eq} = 0.9375\ \Omega$.

(d) The maximum current is $I = 12/0.9375 = 12.8\text{ A}$, below 15 A, so the 15 A fuse is suitable.

Question 9

Two small metal spheres are attached to the ends of an insulating rod of length 5.0 cm. They carry charges $+Q$ and $-Q$ respectively. The rod is suspended horizontally between two parallel metal plates A and B, which are connected to an E.H.T. (extra high tension) supply. The rod is parallel to the metal plates when the E.H.T. is off. After the E.H.T. is switched on, an electric field is set up between the plates and the rod is twisted through an angle of 20° as shown in Figure 9.2.



- On Figure 9.2, sketch the electric field lines due to the potential difference across the plates. (2 marks)
- The potential difference across A and B is 5.0 kV and the separation between the metal plates is 10 cm. The force due to the electric field acting on each sphere is 2.0×10^{-5} N. Find
 - the moment acting on the rod as shown in Figure 9.2 due to the electric forces on the charged spheres; (2 marks)
 - the strength of the electric field E due to the potential difference across the metal plates; (2 marks)
 - the magnitude of the charge Q on the spheres. (2 marks)

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Solution	Marks	Remarks
9. (a) Top view	1A 1A	Accept: field lines are not on Figure 9.2 Correct direction (From A to B) Accept: field lines not in contact with the plates Perpendicular to plates, parallel and evenly spaced Accept: 3 or more field lines NOT accept: 2 or more mistakes in the drawing
	2	
(b) (i) $F \times d = (2.0 \times 10^{-5})(0.05 \cos 20^\circ)$ $= 9.396926 \times 10^{-7} \text{ N m} \approx 9.40 \times 10^{-7} \text{ N m}$	1M 1A	1M $F \times d$ Accept: without consider the component / without component d (0.05, 0.05/2, 2×0.05 , 5, 5/2, 2×5) NOT accept: calculation using separation between plates d (0.1, 10)
	2	
(ii) $E = \frac{V}{d}$ $= \frac{5.0 \times 10^3}{0.1}$ $= 50\,000 \text{ V m}^{-1} \text{ or } \text{N C}^{-1} = 50 \text{ kV m}^{-1} \text{ or } \text{kN C}^{-1}$	1M 1A	Accept: $\frac{5 \times 10^3}{0.1}$, $\frac{5 \times 10^3}{1}$, $\frac{5}{1}$, $\frac{5}{0.1}$
	2	
(iii) $E = \frac{F}{Q}$ $Q = \frac{F}{E} = \frac{2.0 \times 10^{-5}}{5.0 \times 10^4}$ $= 4.0 \times 10^{-10} \text{ C}$	1M 1A	e.c.f. from (b)(ii)
	2	

(a) Draw uniform, parallel electric-field lines directed from plate A to plate B.

(b)(i) The turning moment is $\tau = F(0.050 \cos 20^\circ) = 9.40 \times 10^{-7} \text{ N m}$.(ii) $E = V/d = 5.0 \times 10^3 / 0.10 = 5.0 \times 10^4 \text{ V m}^{-1}$.(iii) $Q = F/E = (2.0 \times 10^{-5}) / (5.0 \times 10^4) = 4.0 \times 10^{-10} \text{ C}$.

Question 10

Given: mass of proton = 1.0073 u mass of α particle = 4.0015 u mass of $^{14}_7\text{N}$ nucleus = 13.9993 u mass of $^{17}_8\text{O}$ nucleus = 16.9947 u When a stationary $^{14}_7\text{N}$ nucleus is bombarded by an α particle, the following nuclear reaction can be triggered, with products $^{17}_8\text{O}$ and X flying off: $\alpha + ^{14}_7\text{N} \rightarrow ^{17}_8\text{O} + \text{X}$

- (a) What is X? (1 mark)
- (b) Based on energy consideration, estimate the minimum kinetic energy, in MeV, of the α particle required for such a nuclear reaction to occur. (2 marks)
- (c) However, when conservation of momentum is also taken into account, the α particle must possess a kinetic energy greater than that found in (b) to bring about such a reaction. Explain. (2 marks)

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Solution	Marks	Remarks
10. (a) proton / ${}^1_1\text{H}$ / p / hydrogen nucleus	1A	Accept: proton / ${}^1_1\text{H}$ / p / 1_1p / H / hydrogen nucleus / hydrogen / hydrogen atom NOT accept: H_2
	1	
(b) Change in mass $\Delta m = (16.9947 + 1.0073) - (13.9993 + 4.0015)$ $= 0.0012 \text{ u}$ Energy $= 0.0012 \times 931$ $= 1.1172 \text{ (MeV)} \approx 1.12 \text{ (MeV)}$ (Or $= 0.0012 \times (1.661 \times 10^{-27}) \times (3.00 \times 10^8)^2$ $= 1.79388 \times 10^{-13} \text{ J} \approx 1.79 \times 10^{-19} \text{ MJ}$)	1M	1M for Δm
	1A	Accept 1.10 to 1.12 (MeV) Answer should be in MeV, not J. MeV could be omitted [Energy $= 0.0012 \times (1.661 \times 10^{-27}) \times (3.00 \times 10^8)^2$ $= 1.79388 \times 10^{-13} \text{ J}$ $\approx 1.79 \times 10^{-19} \text{ MJ}$ $\approx 1.12 \text{ (MeV)}$] Accept (1.79 to 1.80) $\times 10^{-19} \text{ MJ}$
	2	
(c) By the conservation of momentum, as before the reaction occurs the α particle has momentum, the total momentum of the products (= momentum of the α particle) must also be non-zero, i.e. the total KE of the products must greater than 0, thus the α particle should have a larger KE.	1A	1 st 1A should mention the product(s) has/have momentum
	1A	2 nd 1A KE/speed of the products must greater than zero Accept: <u>kinetic</u> energy shared by other particles (NOT accept :energy shared.) NOT accept: against repulsion of nucleus / inelastic collisions / KE loss / consider binding energy
		[The kinetic energy of the α particle = energy calculated in (b) + total KE of the products]
	2	

(a) Particle X is a proton, ${}^1_1\text{H}$.**(b) $\Delta m = (16.9947 + 1.0073) - (13.9993 + 4.0015) = 0.0012 \text{ u}$. Therefore the energy released is $0.0012(931) = 1.12 \text{ MeV}$.****(c) The incoming α -particle has non-zero momentum. Momentum conservation requires the products to have non-zero total momentum and hence kinetic energy, so the incident α -particle must supply both the**

reaction threshold and the products' kinetic energy; its minimum energy is therefore greater than 1.12 MeV.